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Depth determination is a fundamental task for a hydrographer, which requires specific knowledge of the techniques and equipment involved. According to the IHO publication S-44 5th Edition, the uncertainty of depth measurements shall be estimated at the 95% confidence level for every sounding in the survey.

During the last decade, hydrographic surveying has experienced a conceptual change in depth measurement, moving from single-beam echo sounders to multibeam and LiDAR systems. The 2nd generation of swath systems delivers dense coverage of the seafloor with reduced survey time and improved accuracy.

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